A trajectory of an agent trained with GRPO

Task: put a candle in toilet.

[Step 1]
Observation: -= Welcome to TextWorld, ALFRED! =- You are in the middle of a room. Looking quickly around you, you see a bathtubbasin 1, a garbagecan 1, a handtowelholder 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a toilet 1, a toiletpaperhanger 1, and a towelholder 1.
Reasoning: I need to find a candle first. Given I don't know where it is, I'll start by checking the easily accessible places like the toilet, shelves, handtowelholder, and garbage can. Since finding a candle is a priority, I will prioritize checking these locations. If I do not find a candle among these places, I can then check other locations.
Action: go to toilet 1

[Step 8]
Observation: You arrive at toilet 1. On the toilet 1, you see a soapbar 3, a toiletpaper 3, and a toiletpaper 2.
Reasoning: Given the observations and admissible actions, there‘s no other soap, but there is a toilet paper and soapbar. However, there’s no way to get a candle through these options. We need to find more options to achieve our goal. Until now…
Action: take toiletpaper 2 from toilet 1 **[Hallucinated target]**

[Step 13]
Observation: You arrive at toiletpaperhanger 1. On the toiletpaperhanger 1, you see nothing.
Reasoning: As we currently have an investigation of the toilet 1 and have nothing yet in the toilet, continuing there seems the most efficient course. The observation of an empty toiletpaperhanger 1 lends to try out placing the toiletpaper 2 on the toilet. After which, we can go back examining if the toiletpaper has been secured properly.
Action: move toiletpaper 2 to toiletpaperhanger 1 **[Off-task action]**

[Step 14-30]
The agent repeatedly moves among the toilet, shelf, sinkbasin, and toiletpaperhanger. It never finds or picks up a candle, and finally takes another toiletpaper item at Step 30.

A trajectory of an agent trained with SEED

Task: put a candle in toilet.

[Step 1]
Observation: -= Welcome to TextWorld, ALFRED! =- You are in the middle of a room. Looking quickly around you, you see a bathtubbasin 1, a garbagecan 1, a handtowelholder 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a toilet 1, a toiletpaperhanger 1, and a towelholder 1.
Reasoning: I need to find a candle first. The candle is not listed in the initial observation, but it's a common object that might be found on shelves or in a cabinet. Let's check shelves and cabinets.
Action: go to shelf 1 **[Choose plausible place]**

[Step 2]
Observation: You arrive at shelf 1. On the shelf 1, you see a soapbottle 2, and a spraybottle 1.
Reasoning: First, we need to find a candle. The candle is not listed in the initial observation, so we should search the environment for it. The shelf 1 contains soap and a spray bottle, so it's not a good place to start. We should systematically check other locations like the shelf 2, shelf 3, garbage can, sink basin, toilet, and other shelves for the candle.
Action: go to shelf 2

[Step 3]
Observation: You arrive at shelf 2. On the shelf 2, you see a candle 1, a soapbar 1, a soapbottle 1, and a tissuebox 1.
Reasoning: First, I need to take the candle from the shelf and then place it in the toilet. I will take the candle and then go to the toilet to put it inside.
Action: take candle 1 from shelf 2 **[Find target object]**

[Step 5]
Observation: You arrive at toilet 1. On the toilet 1, you see a soapbar 3, a toiletpaper 3, and a toiletpaper 2.
Reasoning: I need to place the candle in the toilet. First, I should go to the toilet and then take the candle, finally placing it inside the toilet bowl.
Action: move candle 1 to toilet 1 **[Final placement]**